INFLOW – Progress Report 2

Rohan Upadhyay | CSCI 412: Senior Seminar II | March 25, 2026

GitHub:
https://github.com/RyanX5/INFLOW

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1. Project Overview

INFLOW is an agent-based simulation designed to study how information and misinformation propagate through social networks. The system models individual users as agents with psychological traits that influence how they receive, believe, and share information.

The central goal of the project is to explore the dynamics that allow misinformation to spread rapidly in online environments. The simulation examines how emotional intensity, individual bias, skepticism, and network structure interact to shape information diffusion.

At the midpoint of the project, the simulation engine, agent models, and visualization system were fully implemented. Since then, the focus has shifted toward experimentation and analysis, using the system to generate insights into propagation behavior.

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2. Implementation Status

The core system remains fully implemented as described in the midterm report. The following modules are complete:

Module	Status	Description
simulation/network.py	Complete	Network generation and topology construction
simulation/agent.py	Complete	Agent behavioral model
simulation/information.py	Complete	Information representation
simulation/engine.py	Complete	Simulation execution engine
visualization/	Complete	React + D3 visualization frontend

New Additions

Module	Status	Description
experiment_runner.py	Complete	Automates simulation experiments
topology_comparison.py	Complete	Compares propagation across network types

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3. System Architecture

INFLOW continues to consist of three primary subsystems:

- simulation/ — backend simulation engine

- visualization/ — frontend visualization system
- output/ — generated CSV datasets

Pipeline:

Network Generation → Agent Initialization → Information Injection → Simulation Execution → Data Logging → Visualization

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4. Experimental Framework

Following the midterm, an experimentation framework was introduced to allow systematic testing of the simulation under different conditions.

Example usage:

```
from experiment_runner import run_experiment
run_experiment(num_agents=100, steps=20)
```

This framework automates:

- network creation
- simulation execution
- data collection
- output generation

The script also generates graphs using matplotlib to compare the information spread in high-emotion intensity information vs low-emotion intensity. Refer to

Experimental Results section below for the graph.

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5. Comparing Topologies

I've also added a script that compares different topologies:

topology_comparison.py

Example usage:

```
from topology_comparison import run_topology_comparison

run_topology_comparison(num_trials=100, num_steps=15, num_agents=100)
```

This scripts automates:

- network creation
- simulation execution
- data collection
- output generation

The script enables comparison across different network structures:

- Random (Erdős–Rényi)
- Small-world (Watts–Strogatz)
- Scale-free (Barabási–Albert)

The script also generates graphs using matplotlib to compare the information spread in different network topologies. Refer to Experimental Results section

below for the graph.

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6. Metrics and Data Logging

The simulation logs propagation data into CSV files, including:

- agent_states.csv
- edges.csv
- info_items.csv
- spread_log.csv

Additional derived metrics include:

Spread fraction:

$$\text{spread fraction} = \frac{\text{agents reached}}{\text{total agents}}$$

Belief statistics:

- average belief
- minimum belief
- maximum belief

These metrics provide deeper insight into how information spreads and how agent beliefs evolve.

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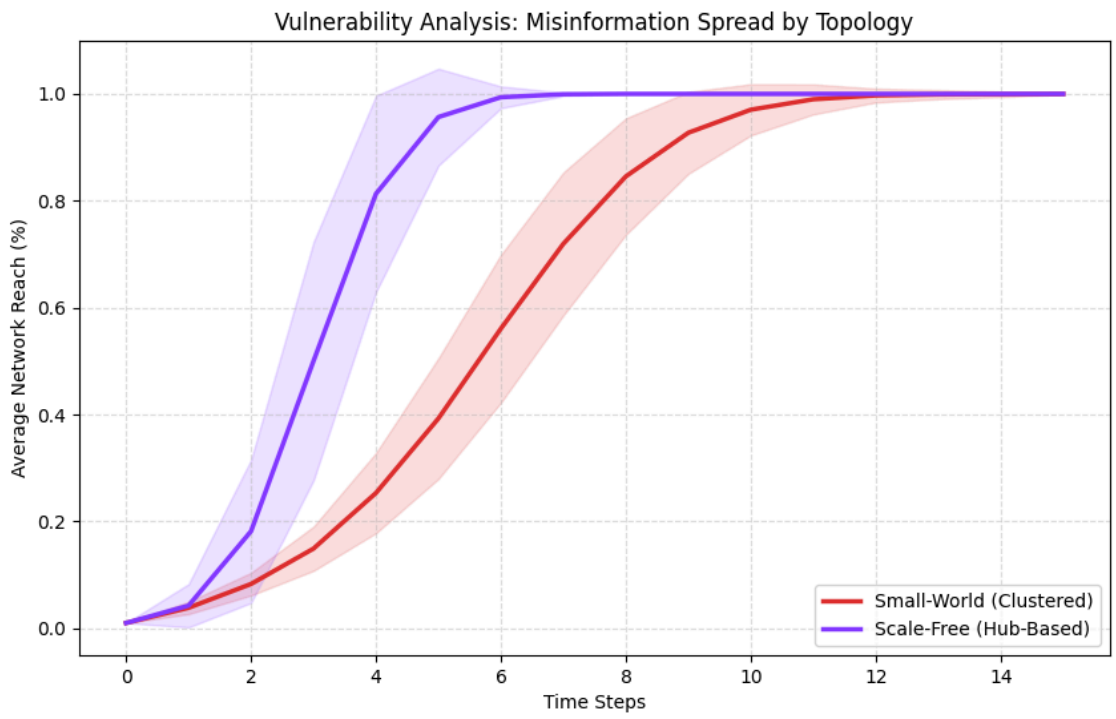
7. Experimental Results

Topology Comparison



Experiments show that network structure has a strong influence on how information propagates through the system.

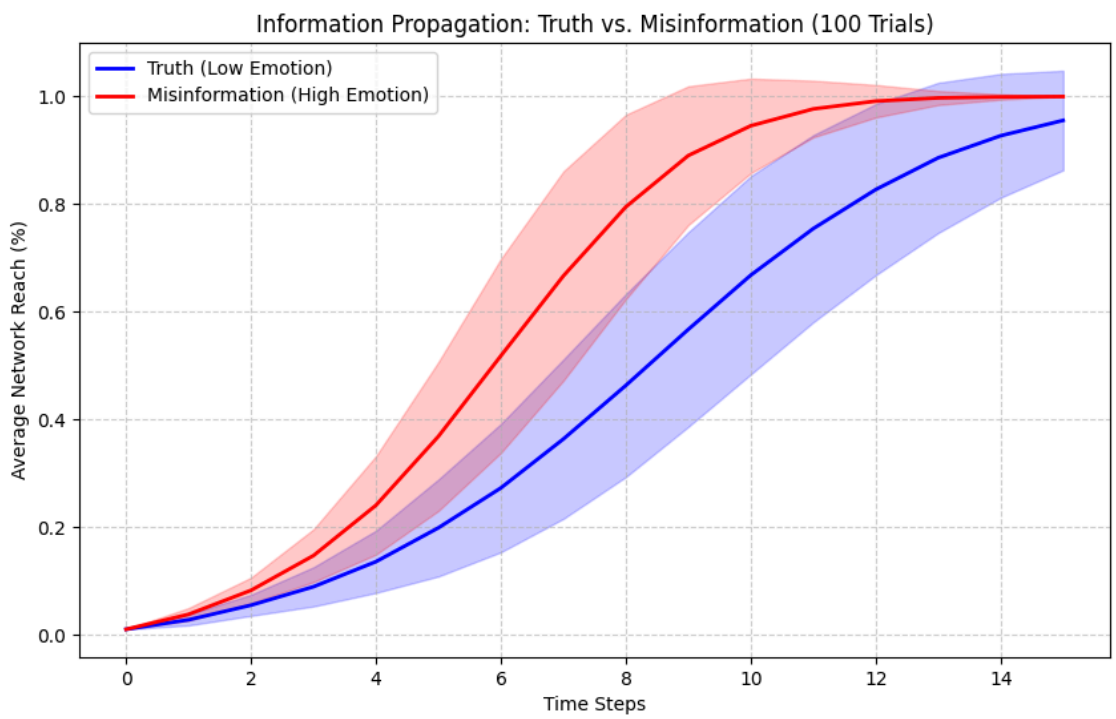
As shown in the figure above, the scale-free network (purple curve) reaches near-total network coverage significantly faster than the small-world network (red curve). The steep initial rise of the scale-free curve indicates rapid early diffusion, driven by highly connected hub nodes.

In contrast, the small-world network demonstrates a more gradual increase in spread, reflecting its clustered structure. Information spreads locally within clusters before reaching the broader network, resulting in delayed saturation.

The shaded regions in the figure further highlight variability across runs, showing that scale-free networks not only spread faster but also exhibit more consistent high reach.

These differences highlight how structural properties of a network can significantly impact the speed and reach of information diffusion.

Propagation Dynamics



The propagation dynamics illustrate how information spreads over time under different conditions.

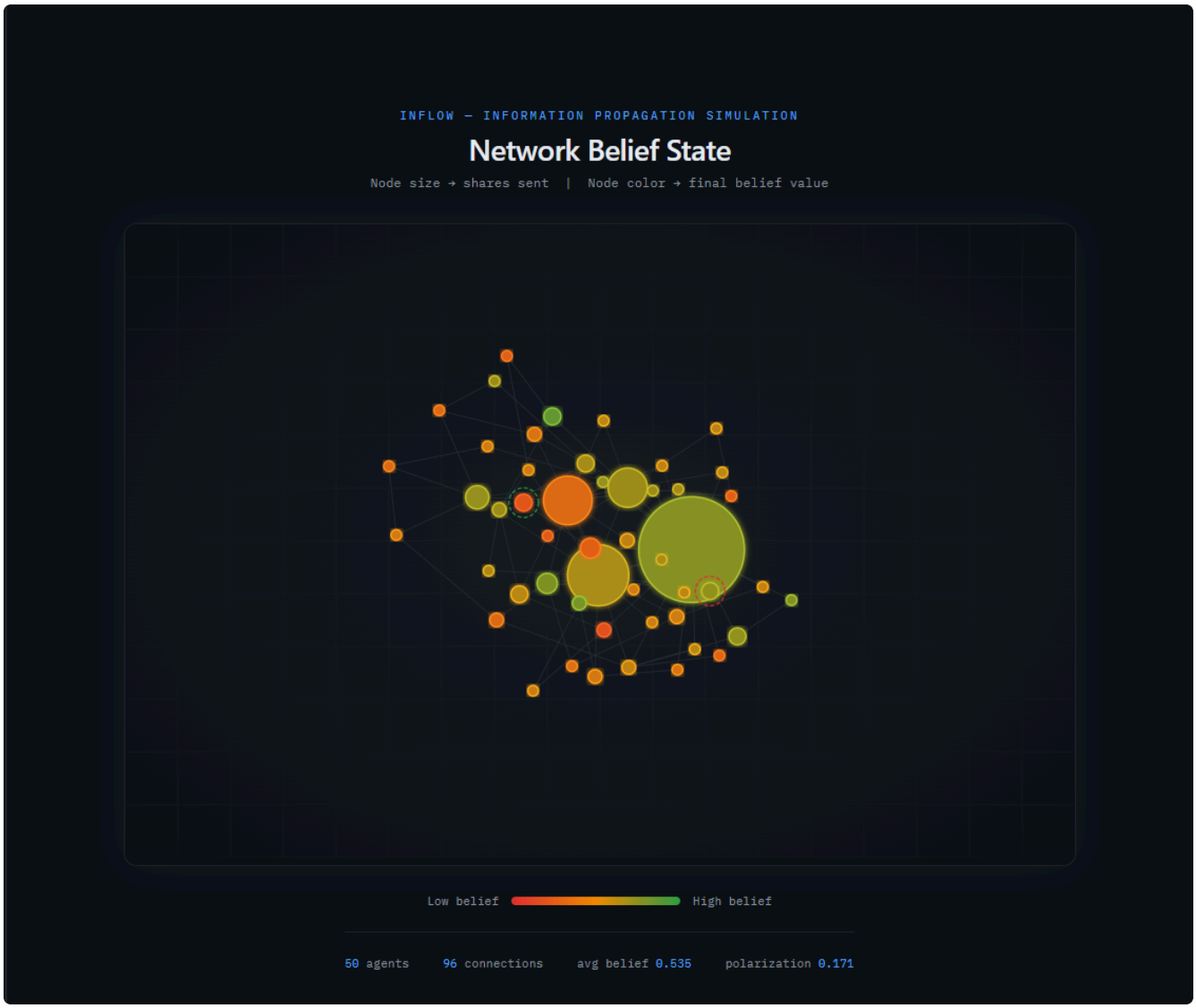
In the figure above, the graph shows a clear upward trend in network reach, with noticeable curvature indicating non-linear growth. Rather than spreading at a

constant rate, the diffusion accelerates over time.

This pattern suggests that once a critical mass of agents becomes influenced, the spread transitions into a rapid growth phase. The smooth but accelerating curve reflects the cumulative effect of repeated sharing and increasing exposure.

The visualization supports the observation that emotionally intense or engaging information tends to produce faster and more widespread propagation.

Cascade Behavior



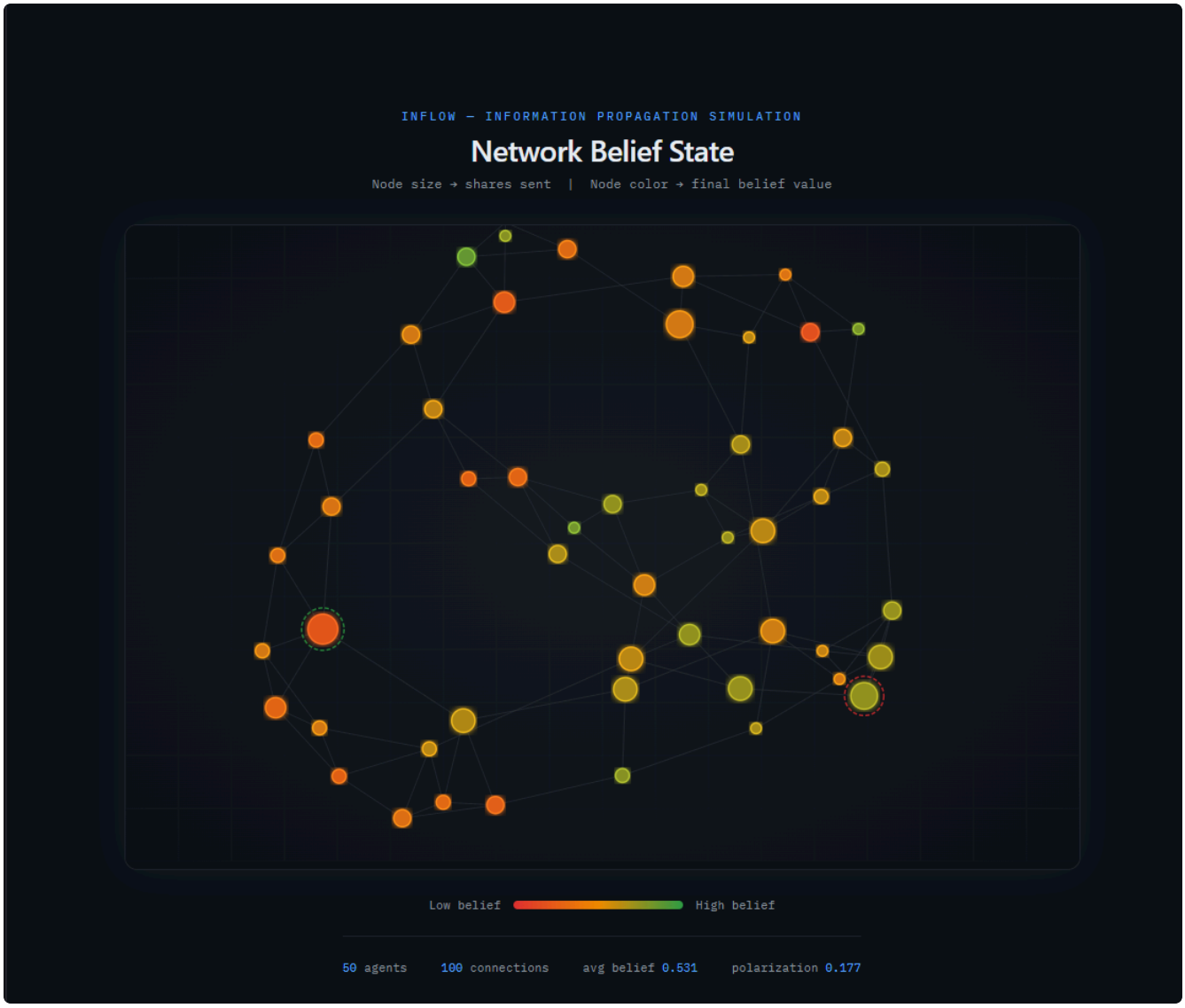
Cascade behavior is particularly evident in the network visualization.

In the above figure, node size represents the number of shares sent, while node color represents belief intensity. The larger nodes clustered in certain regions indicate high-activity agents, many of which act as hubs.

The highlighted nodes (with dashed outlines) demonstrate how influence can originate or concentrate around specific points. Once these nodes become active, they rapidly distribute information to many neighbors, triggering cascade events.

The structure of the network clearly shows how a small number of influential nodes can disproportionately impact overall spread, leading to sudden jumps in network reach.

Belief Evolution



The belief evolution visualization provides insight into how agent beliefs change across the network.

In the figure above, node color ranges from low belief (red) to high belief (green), with intermediate values shown in yellow. The presence of distinct color groupings indicates variation in belief states across the network.

Some regions show clusters of similarly colored nodes, suggesting local convergence of beliefs due to repeated interactions. At the same time, the coexistence of different color regions highlights partial polarization, where different parts of the network settle on different belief levels.

The variation in node sizes also indicates differences in influence, suggesting that more active agents can play a larger role in shaping belief distributions.

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8. Analysis and Insights

The results suggest that:

- Emotional intensity strongly influences spread speed
- Network structure plays a critical role in diffusion patterns
- Scale-free networks are particularly vulnerable to rapid misinformation spread
- Agent interactions produce emergent behavior at the system level

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9. Timeline Update

Original timeline:

Weeks 1–9: Core system development
Week 10: Midterm report

Current status:

- Core system completed by midterm
- Post-midterm work focused on experimentation and analysis

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10. Remaining Work

Remaining tasks include:

- expanding experimental scenarios
- refining model parameters
- improving visualization features
- performing deeper statistical analysis

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11. Challenges Encountered

Challenges included designing meaningful experiments and interpreting complex simulation outputs. Small parameter changes often resulted in significantly different behaviors, requiring careful tuning.

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12. Conclusion

INFLOW has evolved from a simulation system into an experimental platform for studying information propagation. The addition of experimentation tools enables systematic analysis of how network structure and agent behavior affect diffusion dynamics.

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13. Use of AI Tools

AI tools were used for formatting and minor assistance. All implementation and design decisions were made independently.

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14. References

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Watts, D. J., & Strogatz, S. H. (1998). Collective dynamics of “small-world” networks. *Nature*, 393(6684), 440–442.

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